

7.2 Transverse Waves: EM Spectrum & Polarisation

Question Paper

Course	CIE A Level Physics
Section	7. Waves
Topic	7.2 Transverse Waves: EM Spectrum & Polarisation
Difficulty	Medium

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

A beam of vertically polarised monochromatic light is incident on a polarising filter, as shown in Fig. 5.1.

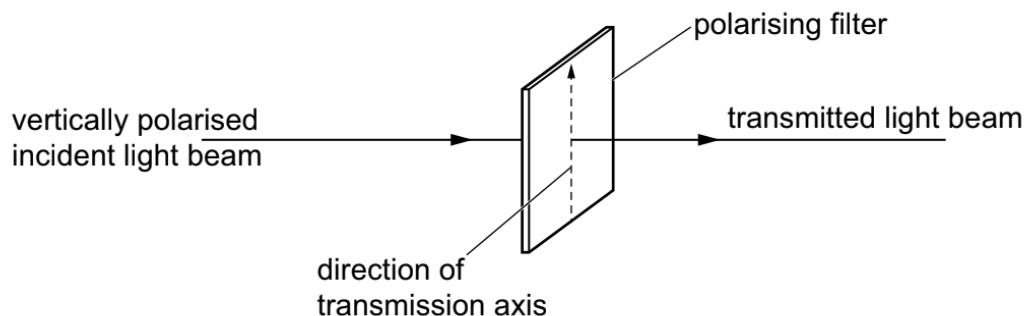


Fig. 5.1

The transmission axis of the filter is initially vertical and the transmitted light beam has the same intensity as the incident light beam.

The filter may be rotated about the direction of the light beam to change the angle of the transmission axis to the vertical.

State one angle of the transmission axis to the vertical that results in no transmitted light beam.

angle = °

[1 mark]

Question 1b

The filter is now positioned with its transmission axis at angle θ to the vertical, as shown in Fig. 5.2.

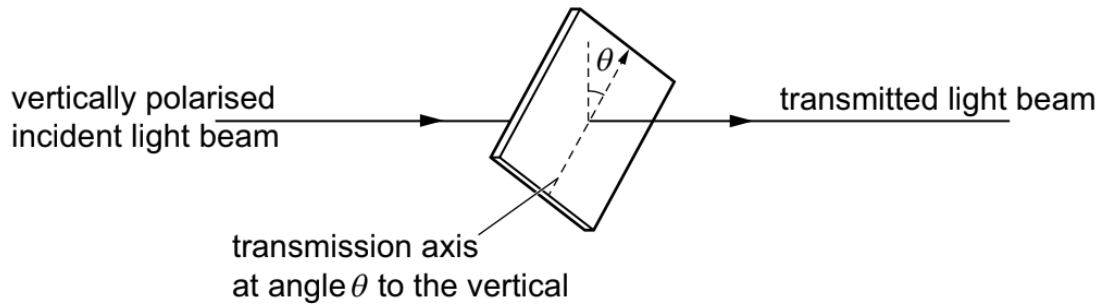


Fig. 5.2

The ratio $\frac{\text{intensity of transmitted light}}{\text{intensity of incident light}}$ is equal to 0.75.

- i)
Calculate angle θ .

$\theta = \dots\dots\dots^\circ$ [2]

- ii)
Calculate the ratio

$$\frac{\text{amplitude of transmitted light}}{\text{amplitude of incident light}}.$$

ratio = $\dots\dots\dots$ [2]

[4 marks]

Question 2a

Unpolarised light is passed through a pair of polarising sunglasses as shown in Fig. 1.1.

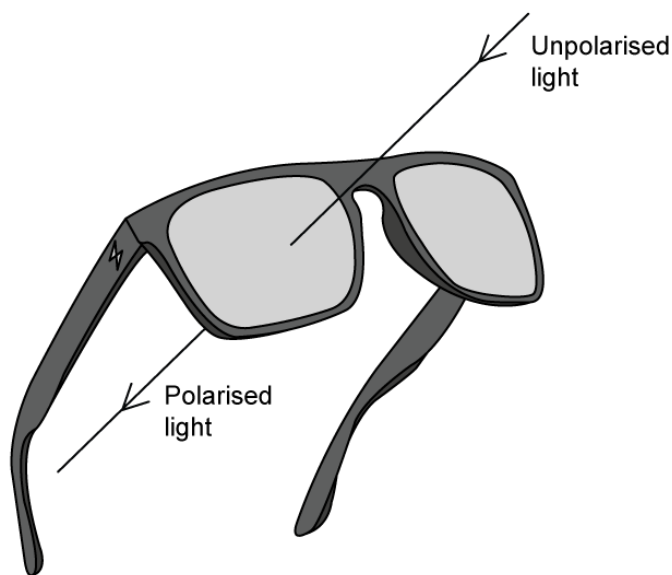


Fig.1.1

The light becomes plane polarised.

Explain the difference between unpolarised and plane polarised light.

[3 marks]

Question 2b

In an experiment to investigate polarising filters, two square filters are used.

Initially, they are placed between the source of unpolarised light and a hand-held digital light meter, as shown in Fig. 1.2. The planes of polarisation of the filters are indicated by double-headed arrows.

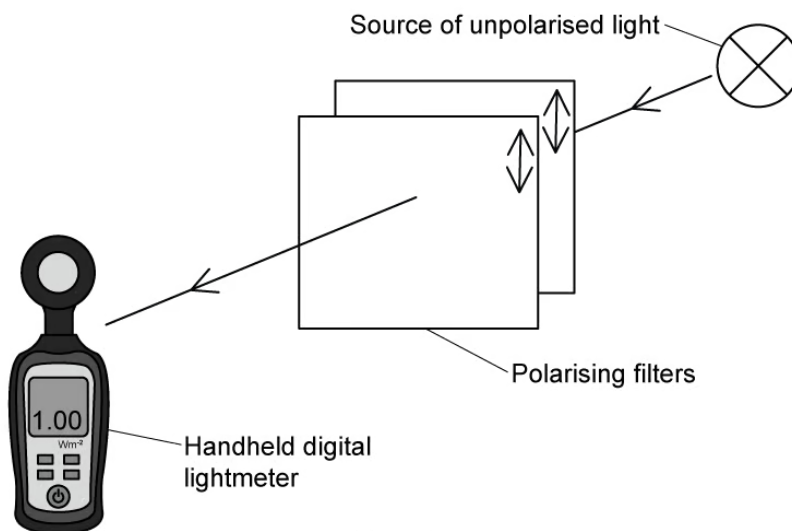
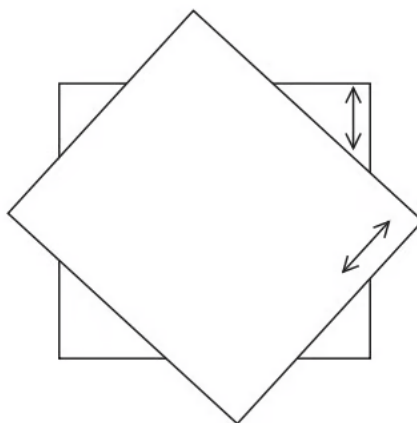


Fig. 1.2

The light meter measures the intensity of radiation as 1.00 W m^{-2} .

One of the polarising filters is then rotated, as shown.



Sketch a graph using the axes as shown in Fig. 1.3. to show how the intensity of light measured by the light meter varies as the filter is rotated from 0° to 90° .



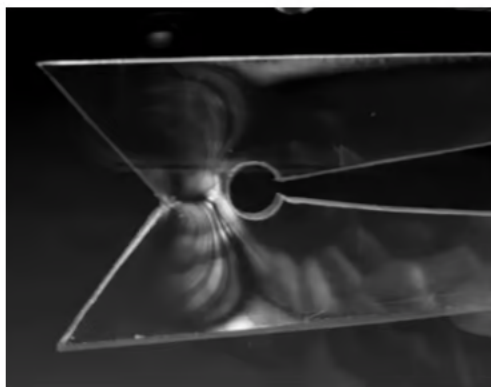
Fig. 1.3

[3 marks]

Question 2c

Polarisation can be used to identify when some materials are under stress. Placing the material under stress causes the plane of polarisation of the light passing through the material to be rotated. The greater the stress, the greater the rotation of the plane of polarisation of the light.

A piece of transparent material under stress is positioned between two polarising filters. There is an angle of 90° between the planes of polarisation of the two polarising filters.



Explain how this photograph can be used to identify areas of the material with different amounts of stress.

[3 marks]

Question 3a

Fig. 1.1. shows two ways in which a wave can be made to travel along a slinky spring.

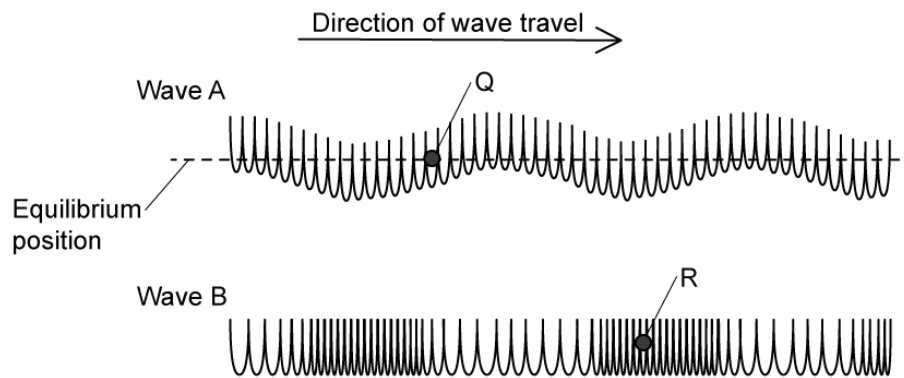


Fig. 1.1

Use arrows to show how points Q and R will move next as the two waves move to the right.

[2 marks]

Question 3b

Use the diagram in Fig. 1.1.

i)

State which wave in Fig. 1.1 can be used to model an electromagnetic wave.

[1]

ii)

Hence explain why it is important to correctly align a radio antenna in order to receive the strongest signal.

[2]

[3 marks]

Question 3c

Light from a filament lamp is viewed through two polarising filters **A** and **B** as shown in Fig. 1.2.

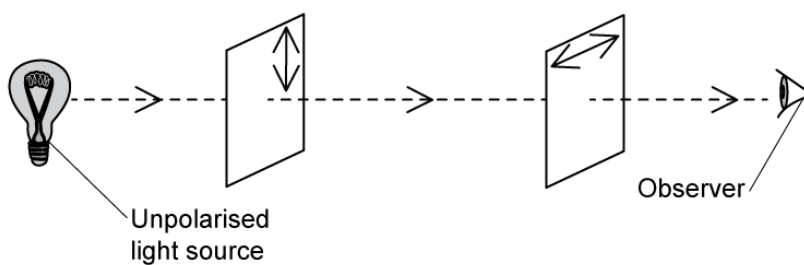


Fig. 1.2

The direction of plane polarisation is indicated on each filter using a double-headed arrow.

Explain why the observer cannot see light from the filament lamp.

[2 marks]

Question 3d

Polaroid sunglasses are sold as sporting equipment to people who fish, sail and kayak.



Fig. 1.2

Explain how polaroid sunglasses can be used to more clearly see objects such as fish, rocks and other obstacles which are under the surface of water, by a person who is above the water.

[3 marks]



Model Answers: Medium

1a

a) No light will be transmitted through the polarising filter if the transmission axis is rotated to be:

- 90° (to the vertical); [1 mark]

[Total: 1 mark]

1b

b)

i) Calculate angle θ :

List the known quantities:

- $$\frac{\text{intensity of transmitted light}}{\text{intensity of incident light}} = \frac{I}{I_0} = 0.75$$

Use Malus's Law to obtain a value for the ratio $\frac{I}{I_0}$:

- $I = I_0 \cos^2 \theta$
- $\frac{I}{I_0} = \cos^2 \theta = 0.75$ [1 mark]

Calculate angle θ :

- $\cos \theta = \sqrt{0.75}$
- $\theta = \cos^{-1}(\sqrt{0.75}) = 30^\circ$ [1 mark]

ii) Calculate the ratio $\frac{A}{A_0}$:

- $I \propto A^2$ [1 mark]
- $\frac{I}{I_0} \propto \left(\frac{A}{A_0}\right)^2$
- $0.75 \propto \left(\frac{A}{A_0}\right)^2$
- $0.75^{\frac{1}{2}} \propto \frac{A}{A_0}$
- Therefore, $\frac{A}{A_0} = 0.866 = 0.87$ (2 s.f.) [1 mark]

[Total: 4 marks]

- Linking **Malus's Law** to the **ratio given** is the key to answering part (i)

2a

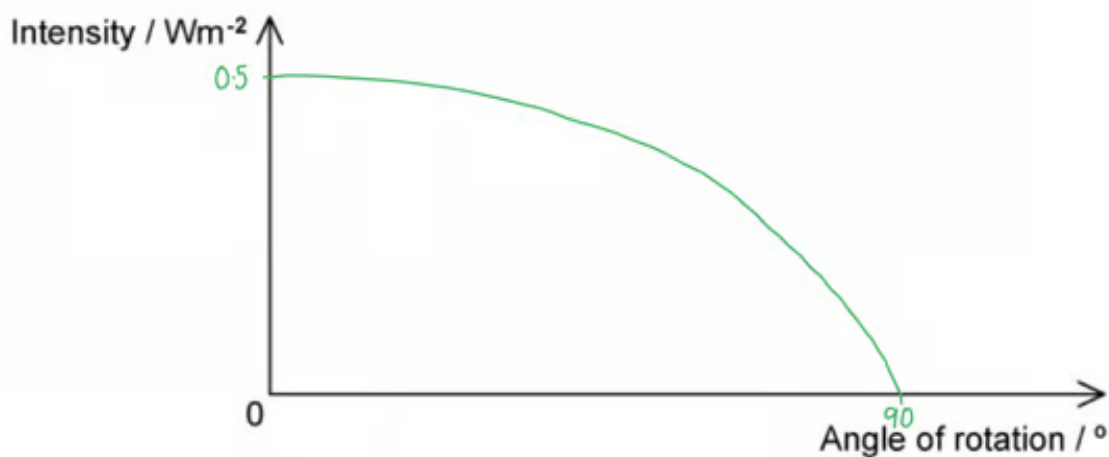
a) Explain the difference between unpolarised and plane polarised light:

- Unpolarised light vibrates/oscillates in all planes; [1 mark]
- Plane polarised light vibrates/oscillates in one plane; [1 mark]
- (Planes of oscillation are) perpendicular to the direction of wave travel; [1 mark]

[Total: 3 marks]

2b

b) Sketch a graph to show how the intensity of light measured by the light meter varies as the filter is rotated from 0° to 90° :



- Correct numbers added to both axes; [1 mark]
- Single maximum at 0° and single minimum at 90° [1 mark]
- Graph starts at (0, 0.5) **AND** gradient increases to (90, 0); [1 mark]

[Total: 3 marks]

2c

c) Explain how this photograph can be used to identify areas of the material with different amounts of stress:

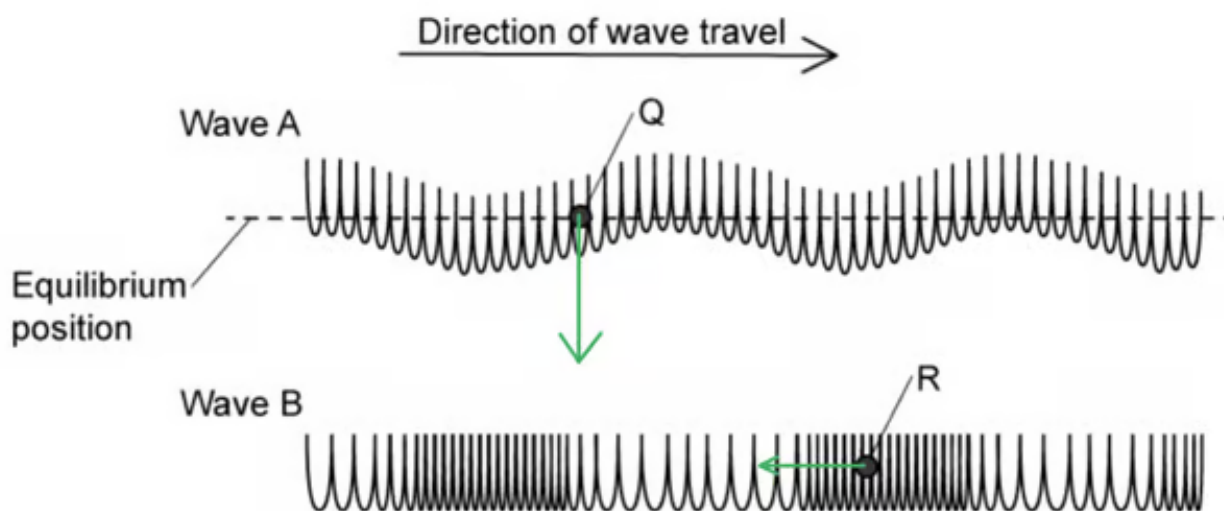
- (Polarising) filters at 90° to each other do not allow light to pass through; [1 mark]
- Rotation of plane of polarisation (due to stress) allows light to pass; [1 mark]
- Darker areas represent less stress **OR** brighter areas represent greater stress; [1 mark]

[Total: 3 marks]

3a

a) Use arrows to show how points Q and R will move next as the two waves move to the right:

Correctly drawn arrows look like:



- Q moving downwards; [1 mark]
- R moving to the left; [1 mark]

[Total: 2 marks]

You may have thought that R will follow the direction of the wave. However, it is to the left of the centre of a compression. Therefore as the compression expands to a rarefaction, R will be pushed backwards compared to the direction of travel of the wave.

3b

b) Use the diagram in Fig. 1.1.

i) State which wave can be used to model an electromagnetic wave:

- Wave A can be used to model electromagnetic waves; [1 mark]

ii) Hence explain why it is important to correctly align a radio antenna in order to receive the strongest signal.

- Transmitted radio waves are polarised; [1 mark]
- Antennae must be aligned to the same angle / plane of polarisation; [1 mark]

[Total: 3 marks]

3c

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3b

b) Use the diagram in Fig. 1.1.

i) State which wave can be used to model an electromagnetic wave:

- Wave A can be used to model electromagnetic waves; [1 mark]

ii) Hence explain why it is important to correctly align a radio antenna in order to receive the strongest signal.

- Transmitted radio waves are polarised; [1 mark]
- Antennae must be aligned to the same angle / plane of polarisation; [1 mark]

[Total: 3 marks]

3c

c) Explain why the observer cannot see light from the filament lamp:

- Light passing through polarising filter **A** is vertically polarised; [1 mark]
- Only light in the horizontal plane / horizontally polarised light will pass through polarising filter **B**; [1 mark]

[Total: 2 marks]

3d

d) Explain how polaroid sunglasses can be used to more clearly see objects which are under the surface of water, by a person who is above the water:

- Minimise / reduce glare from sunlight; [1 mark]
- Reflected light (from the surface of the water) undergoes partial plane polarisation; [1 mark]
- Glasses' polarising filter absorbs light in one plane of polarisation (therefore reducing glare; [1 mark]

[Total: 3 marks]

